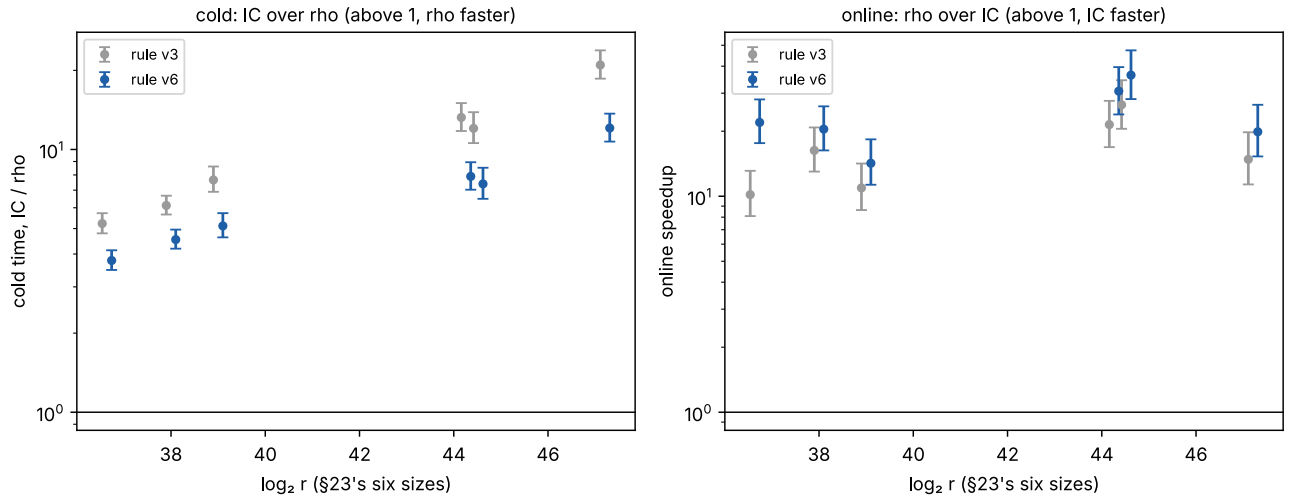
The rule's comparison at baseline v6

One unseen public point a process, the index calculus against Pollard rho on that same point, at ledger §23′s six sizes, 64 targets and four A/A repeats a size: 408 rows, every one checked, every logarithm §23′s. v6 is v4 (main's `92246530`) with R06′s scan key and R08′s folding subtraction. Run on 2026-10-16 on an Emerald Rapids host of the reference class. Protocol and amendment 1: `research/ic_tool_program/rule/v6/PROTOCOL.md`.

- **Verdict (§23′s):** online the index calculus is faster than rho at all six sizes (14.2–36.4×). Cold, rho is still faster at all six.
- **Distance remaining:** rho is 3.78–12.06× faster cold (rule v3: 5.23–20.97×). At the three largest sizes it is 7.90, 7.40 and 12.06× (13.24, 12.03, 20.97×). The gap still grows with the group, as r^0.142 [0.105, 0.178] (r^0.176 at v3).
- **Checks:** 408 of 408 rows pass the `vs_rho` claim check (verified_answer_controlled_wall). The pin held. The reference check is admissible at every size: rho at 0.70–0.85 of the strong walk's time a step. Four rows' contended first attempts re-ran clean.
- **Amendment 1:** the claim writer gained the ledger's five newer `vs_rho` fields, and the strong-rho fixture's re-imported size gate (which refused n = 47, 57 and 59) was widened. No row was re-timed.
- **Pricing (AGENTS.md §5a):** both arms are timed in one process, so no native work is unpriced on either arm, and the 0 / 0.1 / 1 GAE band gives one figure. The counted-unit readings stay lower bounds.



research/ic_tool_program/rule/v6/analysis.json and rule/v3/analysis.json; 408 checked rows each; 95% bootstrap intervals over 64 targets

Figure 1: Left: cold time, IC over rho (above 1, rho faster). Right: online speedup, rho over IC. Rule v3 grey, rule v6 blue; 95% bootstrap intervals over 64 targets.

curve	$\log_2$ r	cold, v6 [95%]	cold, v3	online, v6 [95%]	online, v3	rho / strong
`icv1-f2m47-t22705043-f4e44623`	36.64	3.78 [3.48, 4.13]	5.23	21.97 [17.62, 28.07]	10.17	0.76
`icv1-f2m57-tm747311035-c1f545af`	38.0	4.54 [4.19, 4.95]	6.12	20.45 [16.3, 26.08]	16.3	0.85
`icv1-f2m41-tm2308219-7f48b14a`	39.0	5.12 [4.63, 5.72]	7.66	14.22 [11.3, 18.34]	10.93	0.75
`icv1-f2m53-tm56619371-dac20a85`	44.26	7.9 [7.03, 8.94]	13.24	30.66 [23.92, 39.61]	21.47	0.83
`icv1-f2m59-tm943548413-98844ecc`	44.52	7.4 [6.49, 8.52]	12.03	36.37 [28.18, 47.37]	26.5	0.7
`icv1-f2m61-t158598901-ab42b6c5`	47.21	12.06 [10.72, 13.69]	20.97	19.88 [15.29, 26.51]	14.82	0.72